

Unit 4 Pre-Test

Angle Relationships & Triangles

Standard assessed: NC.8.G.5

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.5 Understand angle relationships at a point and on a line.

• I can find missing angles using supplementary, complementary, vertical, and linear-pair relationships.

Match each vocabulary term to its definition, then find each missing angle.

D ▾ supplementary angles ✓	A. Two angles whose measures add to 90° .
A ▾ complementary angles ✓	B. Opposite angles formed by two intersecting lines; they are equal.
B ▾ vertical angles ✓	C. A line that crosses two or more other lines.
C ▾ transversal ✓	D. Two angles whose measures add to 180° .
E ▾ corresponding angles ✓	E. Matching-position angles at two intersections; equal when the lines are parallel.
F ▾ exterior angle ✓	F. The angle formed outside a triangle when one side is extended.

Angle facts — reference

Straight line = 180° · Triangle angles sum to 180° · Right angle = 90°

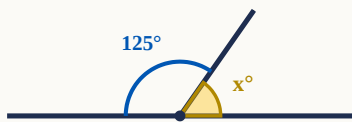
Exterior angle of a triangle = sum of the two remote interior angles.

1. Supplementary

Two angles are supplementary. One is 125° . Find the other.

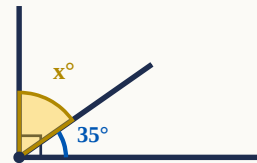
2. Complementary

Two angles are complementary. One is 35° . Find the other.



x =

answer: 55°

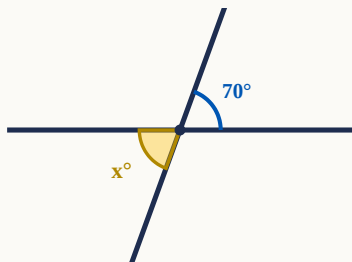


x =

answer: 55°

3. Vertical

Two lines cross. One angle is 70° . Find its vertical angle.

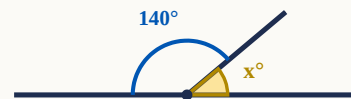


x =

answer: 70°

4. Linear pair

Angles on a straight line are 140° and x. Find x.

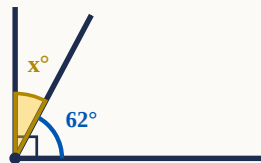


x =

answer: 40°

5. Complementary

Two angles are complementary. One is 62° . Find the other.

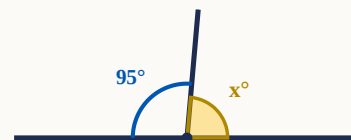


x =

answer: 28°

6. Linear pair

Two angles form a linear pair. One is 95° . Find the other.



x =

answer: 85°

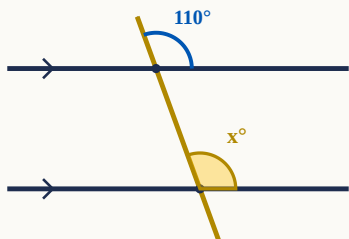
NC.8.G.5 Find angles formed when parallel lines are cut by a transversal.

• I can use corresponding, alternate, and co-interior angle relationships to find unknown angles.

Find each unknown. Where a variable is given, solve for x.

7. Corresponding

Parallel lines cut by a transversal. Find the corresponding angle to 110° .

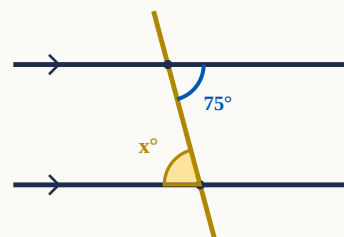


x =

answer: 110°

8. Alt. interior

Alternate interior angles; one is 75° . Find the other.

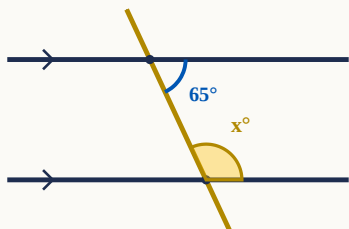


x =

answer: 75°

9. Co-interior

Same-side (co-interior) angles; one is 65° . Find the other.

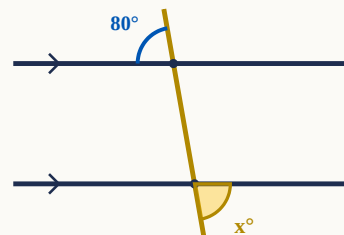


x =

answer: 115°

10. Alt. exterior

A transversal makes an 80° angle. Find its alternate exterior angle.



x =

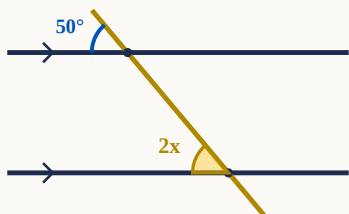
answer: 80°

11. Solve

Angles $2x$ and 50° are corresponding. Find x.

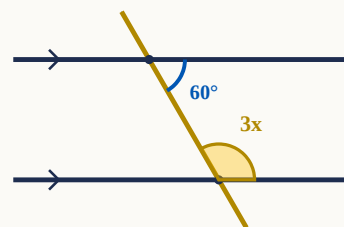
12. Solve

Co-interior angles $3x$ and 60° add to 180° . Find x.



$x =$

answer: $x = 25$



$x =$

answer: $x = 40$

SCORE 5.0 Advanced Content

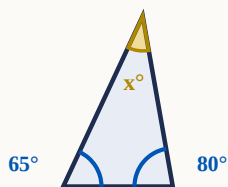
NC.8.G.5 Apply the triangle angle-sum and exterior-angle theorems.

• I can use the triangle angle sum and exterior-angle relationship to solve multi-step problems.

Find each unknown angle or value.

13. Triangle sum

A triangle has angles 65° and 80° .
Find the third angle.

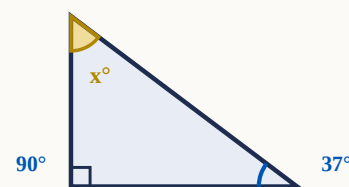


$x =$

answer: 35°

14. Triangle sum

A triangle has angles 90° and 37° .
Find the third angle.



$x =$

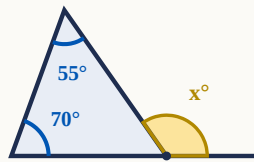
answer: 53°

15. Exterior angle

Remote interior angles are 55° and 70° . Find the exterior angle.

16. Exterior angle

An exterior angle is 120° ; one remote interior is 45° . Find the other remote.

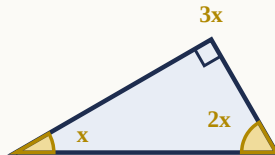


$x =$

answer: 125°

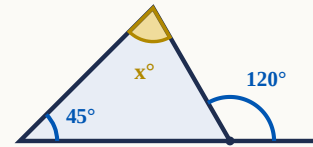
17. Solve

A triangle has angles x , $2x$, and $3x$.
Find x .



$x =$

answer: $x = 30$

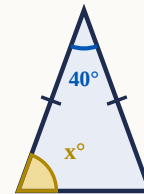


$x =$

answer: 75°

18. Isosceles

An isosceles triangle has a vertex angle of 40° . Find one base angle.



$x =$

answer: 70°